

# Network Science

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## Lecture 03: Strong and Weak Ties

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# From Graphs to Social Structure

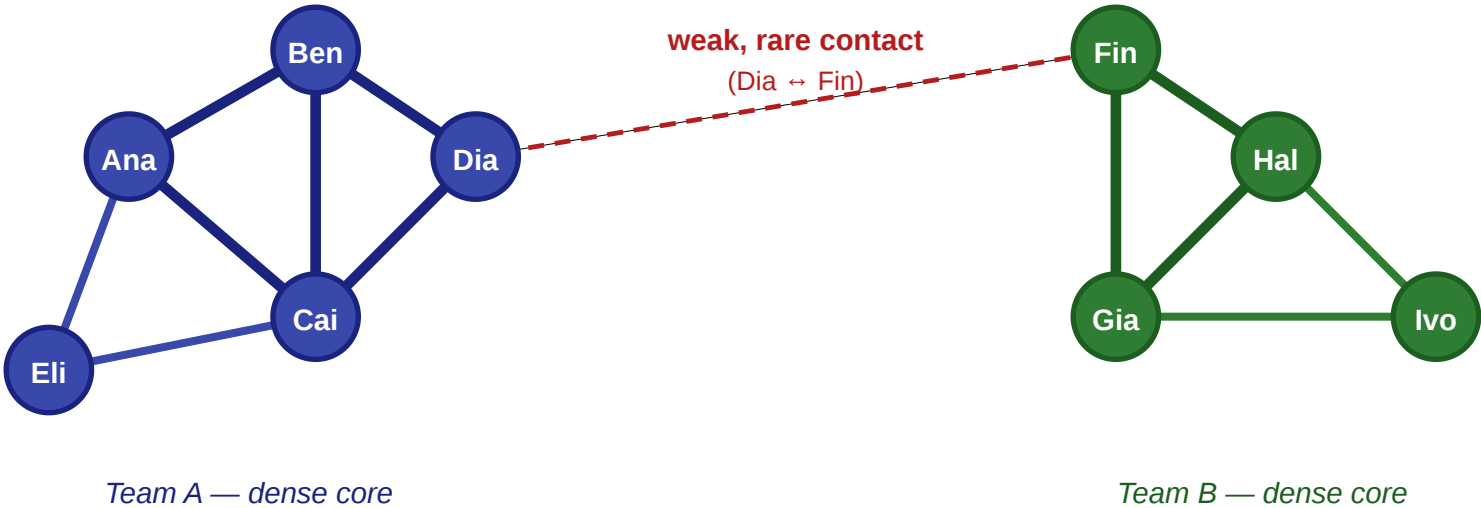
Last time we built the formal language of graphs. Today we use it to ask a different kind of question: once we have the graph, how do we explain what happens on it?

Consider a small workplace.

# A Motivating Scenario

## A small workplace: five questions in one picture

nodes = employees · thick edges = close contacts · thin edges = acquaintances



### Questions we'd like to answer from this picture

- ① Will Eli and Ben become friends? ② How do we separate close from distant ties?  
③ Which single edge controls all information flow between teams? ④ Where would a rumor spread fastest? ⑤ Which tie would we cut to isolate a leak?

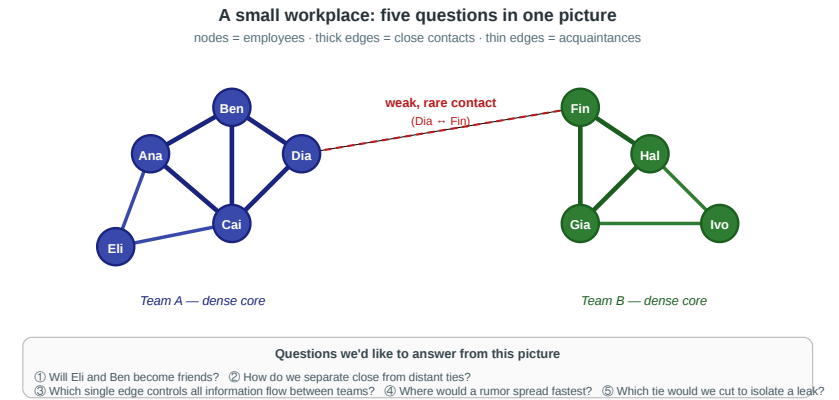
Two teams, each tightly connected internally, linked only by a rare contact between Dia and Fin. Thick lines are close daily contacts, the thin dashed line is an occasional one.

# Five Questions from One Picture

Looking at the scenario, we can further ask:

1. **Edge formation.** Will Eli and Ben eventually become friends, given that they share a close contact?
2. **Tie strength.** How do we formally separate the close ties inside each team from the occasional Dia–Fin contact?
3. **Shortcuts.** Which single edge controls all information flow between the two teams?
4. **Diffusion.** Where would a rumor spread fastest — starting from Ana, or from Hal?
5. **Intervention.** Which tie would we cut to isolate a leak between the teams?

Every concept in today's lecture is a formal answer to one of these questions.



# The Catch: Theory vs. Measurement

Each of those five questions has a **theoretical** form and a **measurement** form — and they are not the same.

- The theoretical form asks: *what structure would explain this behavior if we could see everything?*
- The measurement form asks: *what can we actually compute from a real graph where tie strengths, intentions, and labels are hidden?*

Today's lecture lives in the gap between those two.

## Today

- triadic closure as a theory of friendship formation
- typed edges and Strong Triadic Closure as an edge-labeling model
- why the exact labeling is NP-hard — and what to do instead
- structural heuristics: clustering coefficient and neighborhood overlap
- the weak-tie theorem and its empirical test at scale

# In a bit more detail

Every question below has a **theoretical** form and a **measurement** form. The lecture deliberately treats them as distinct tasks.

| Question                              | Theoretical construct             | Measurable proxy               |
|---------------------------------------|-----------------------------------|--------------------------------|
| How do new ties form?                 | Triadic closure                   | Triangle density over time     |
| Are ties of different kinds?          | Strong / weak edge labeling + STC | Clustering coefficient         |
| Which edges carry novel info?         | Local bridge                      | Neighborhood overlap           |
| Why are bridges weak?                 | Weak-Tie Theorem                  | Tie-strength vs. overlap plots |
| Which ties keep the network together? | Robustness under edge removal     | Giant-component size           |

# Takeaway

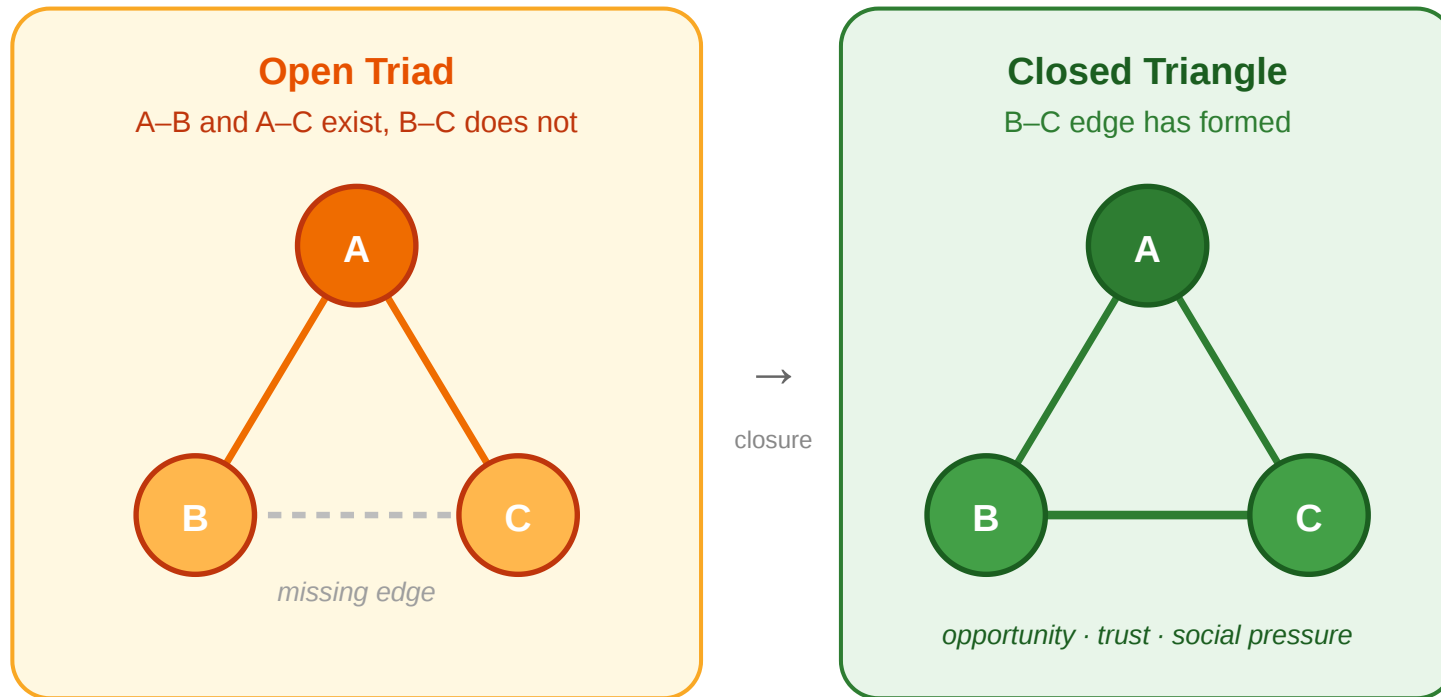
Network science in this lecture is a loop: we build a theoretical model of edge formation and tie strength, discover that exact inference from data is intractable, replace the model with polynomial-time structural proxies, and test whether the proxies match a theoretical prediction at scale.

# Triadic Closure

**Guiding Question:** Why do friends-of-friends so often become linked — and what does that imply for how social graphs grow?

# Open Triads and Closure

**Triadic Closure.** If a node  $A$  has edges to two distinct nodes  $B$  and  $C$ , the probability that an edge  $(B, C)$  forms over time is significantly higher than for two random nodes.



The dashed edge in the open triad is the missing one. Triadic closure predicts it will tend to appear over time, producing the closed triangle on the right.

# Triadic Closure as a Model

Triadic closure is not an observation — it is a claim about how edges appear. As a model, it makes predictions:

- the graph should accumulate triangles over time
- average distances should shrink
- nodes with many neighbors should have increasingly clustered neighborhoods
- long-range ties should become rarer relative to local ones

# Empirical Test: University Email Network

Kossinets & Watts (2006), *Empirical Analysis of an Evolving Social Network*, [Science 311, 88–90](#).

They reconstructed the social network of **43 553 students, faculty, and staff** at a large US university from one year of time-stamped email headers, tracking which pairs started exchanging messages over time.

**Headline result.** Two people who shared **one** mutual email contact were roughly **30 times** more likely to begin emailing each other than two people with no mutual contact. The effect grew monotonically with the number of shared contacts — direct, longitudinal evidence for triadic closure at scale.

# What the Study Could Not See

Kossinets & Watts observed only one signal per edge: *did an email pass between these two people, yes or no?*

- They could not see whether an edge was a daily collaboration or a one-off administrative ping.
- They could not distinguish a pair who spoke for hours off-platform from a pair who had never met.
- Every edge in the dataset was, by construction, **binary and untyped**.

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- They could not distinguish a pair who spoke for hours off-platform from a pair who had never met.
- Every edge in the dataset was, by construction, **binary and untyped**.

Closure is visible in their data. **Tie strength is not**. The next module takes that limitation seriously: if we want to model strong vs. weak ties, we need a formal object — an edge labeling — that the email graph never gave us.

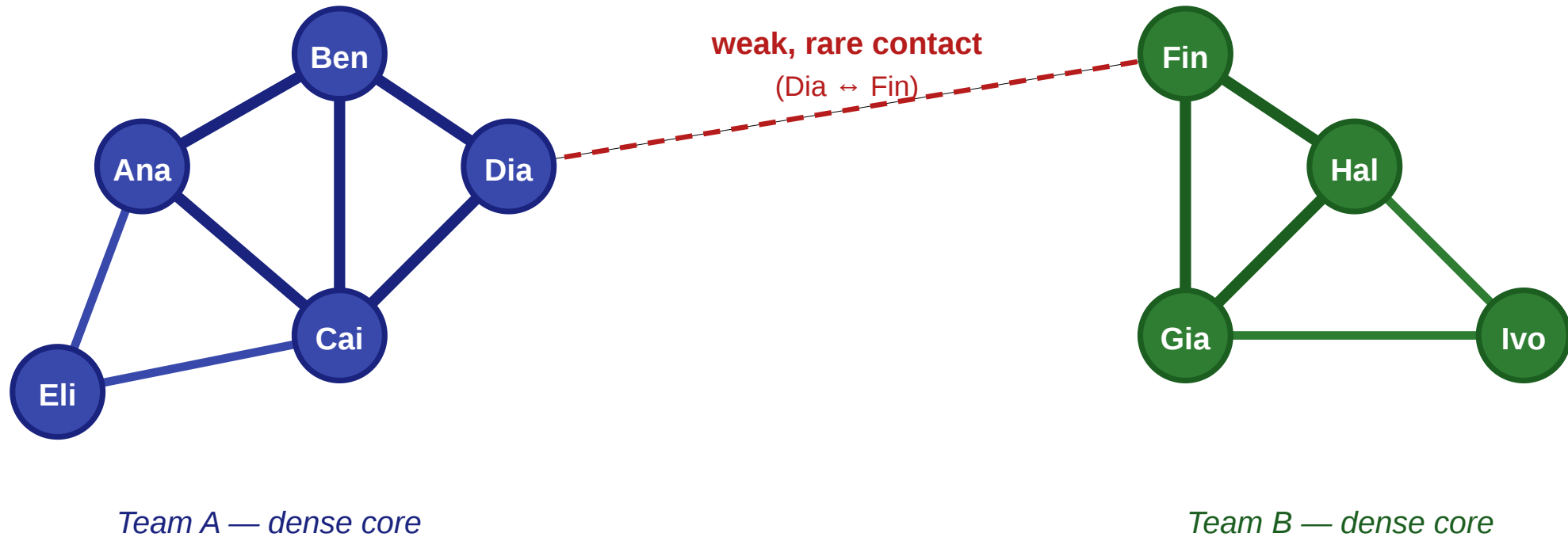
# Takeaway

Triadic closure is a generative theory of edge formation. Kossinets & Watts confirmed it at scale on a university email network — but their binary email data could not distinguish strong from weak ties. To go further, we need to model tie strength explicitly.

# Quick Check

## A small workplace: five questions in one picture

nodes = employees · thick edges = close contacts · thin edges = acquaintances



### Questions we'd like to answer from this picture

- ① Will Eli and Ben become friends?
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# Quick Check — Solution

2. A single failed prediction does not refute a probabilistic model. But *persistent* failure across many such triples would indicate either that the closure mechanism is weak in this population or that some competing mechanism — interpersonal conflict, structural separation, or platform effects — is actively preventing closure.

# Typed Edges and Strong Triadic Closure

**Guiding Question:** If ties come in different *kinds*, what constraint does that place on how they coexist around a node?

# Edges Are Not All the Same

In the scenario, the Ana–Ben edge and the Dia–Fin edge are clearly different:

- Ana–Ben is a daily interaction between people in the same team
- Dia–Fin is a rare, occasional contact between people in different teams

Graph theory's default edge is binary: either present or absent. We now extend it.

**Edge labeling.** A graph  $G = (V, E)$  together with a function

$$\ell : E \rightarrow \text{Strong, Weak}$$

assigning a *type* to every edge. We write  $S(v)$  for the set of strong neighbors of  $v$  under  $\ell$ .

# Strong Triadic Closure as a Constraint on Labelings

**Strong Triadic Closure (STC).** A labeling  $\ell$  of  $G$  satisfies STC if, for every node  $v$  and every pair  $u_1, u_2 \in S(v)$  of strong neighbors of  $v$ , the edge  $(u_1, u_2)$  exists in  $E$  — *with any label* (Strong or Weak). STC constrains which pairs must be **connected**, not how the closing edge is labeled.

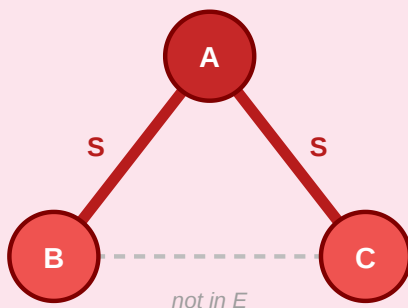
## Strong Triadic Closure as an edge-labeling constraint

$\ell : E \rightarrow \{S, W\}$  · STC: if  $\ell(A,B)=S$  and  $\ell(A,C)=S$  then edge  $(B,C)$  must exist (with any label)

— S = strong (thick)    — W = weak (thin)    - - - edge absent from  $E$

### Invalid labeling (STC violated)

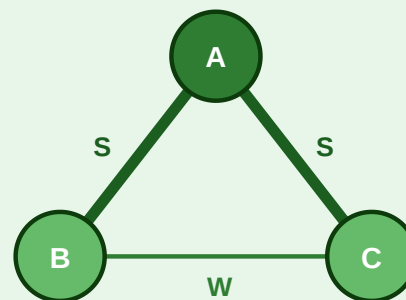
A has two strong ties but the edge  $(B,C)$  is not in  $E$



✗ two strong neighbors of A are not connected

### Valid labeling (STC satisfied)

Add edge  $(B,C)$  to  $E$  — its label can be S or W



✓ every pair of strong neighbors of A is connected (label irrelevant)

STC is a *property of a labeling*, not a property of a graph. The same graph can admit valid STC labelings (right) and invalid ones (left). The question "does the graph satisfy STC?" is ill-posed — the right question is "which labelings of this graph satisfy STC?".

# The Recovery Problem — MaxSTC

In real data we observe the graph  $G$  but not the labeling  $\ell$ . Given  $G$ , we want to recover the labeling that best explains the observed structure — the one with as many strong edges as possible subject to STC.

**MaxSTC.** Given  $G = (V, E)$ , find a labeling  $\ell$  satisfying STC that maximizes  $|e \in E : \ell(e) = \text{Strong}|$ .

# Complexity of MaxSTC

**Result (Sintos & Tsaparas, 2014).** Deciding MaxSTC is **NP-hard** on general graphs. Polynomial-time algorithms exist for certain restricted graph classes (e.g., cographs and several bipartite-like structures).

Consequence: on any realistically sized social network, we **cannot** recover the true STC-maximizing labeling exactly.

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Consequence: on any realistically sized social network, we **cannot** recover the true STC-maximizing labeling exactly.

This is a **modeling problem**, not just an engineering problem. The theoretical construct we care about is computationally out of reach. We need a different strategy.

# What We Do Instead

Since we cannot recover the true labeling, we take a different route:

1. Define structural **heuristics** that approximate the "strong"/"weak" character of edges in polynomial time.
2. Derive a theoretical **prediction** from STC (the weak-tie theorem).
3. **Test** the prediction on real data, using the heuristics as proxies for the unobservable labeling.

This is the plan for the next three modules.

# Takeaway

Tie strength is most precisely modeled as an edge labeling, and Strong Triadic Closure is a constraint on such labelings. Recovering the best labeling is NP-hard — so we must approximate the exact theory with polynomial-time structural heuristics.

# Quick Check

Consider four nodes  $A, B, C, D$  with edges  $A-B, A-C, A-D$ , and  $B-C$ . No other edges exist.

1. Find a labeling of this graph that satisfies STC with the maximum number of strong edges.
2. Is the optimal labeling unique?

# Quick Check — Solution

**1. Optimal labeling.** At node  $A$ , making all three ties strong would force  $B-D$ ,  $C-D$ , and  $B-C$  to exist. Only  $B-C$  exists, so at most *two* of  $A$ 's ties can be strong — and they must be the two whose endpoints are actually connected:  $A-B$  and  $A-C$ . The edge  $B-C$  can be strong without further constraint. So:

$$\ell(A-B) = \ell(A-C) = \ell(B-C) = S, \quad \ell(A-D) = W$$

gives **three** strong edges — the maximum.

**2. Uniqueness.** In this small graph, yes. On larger graphs, MaxSTC can have exponentially many optimal labelings — another reason the problem is hard.

# Structural Heuristics for Tie Strength

**Guiding Question:** Without access to the true edge labels, how can we estimate which parts of a network behave like "strong, embedded" regions and which behave like "weak, bridging" ones?

# Clustering Coefficient

**Local clustering coefficient.** For a node  $i$  with degree  $k_i$ :

$$C_i = \frac{\text{edges among } i\text{'s neighbors}}{\binom{k_i}{2}}$$

$C_i \in [0, 1]$ .  $C_i = 1$  means  $i$ 's neighborhood is a complete clique;  $C_i = 0$  means none of  $i$ 's neighbors are linked.

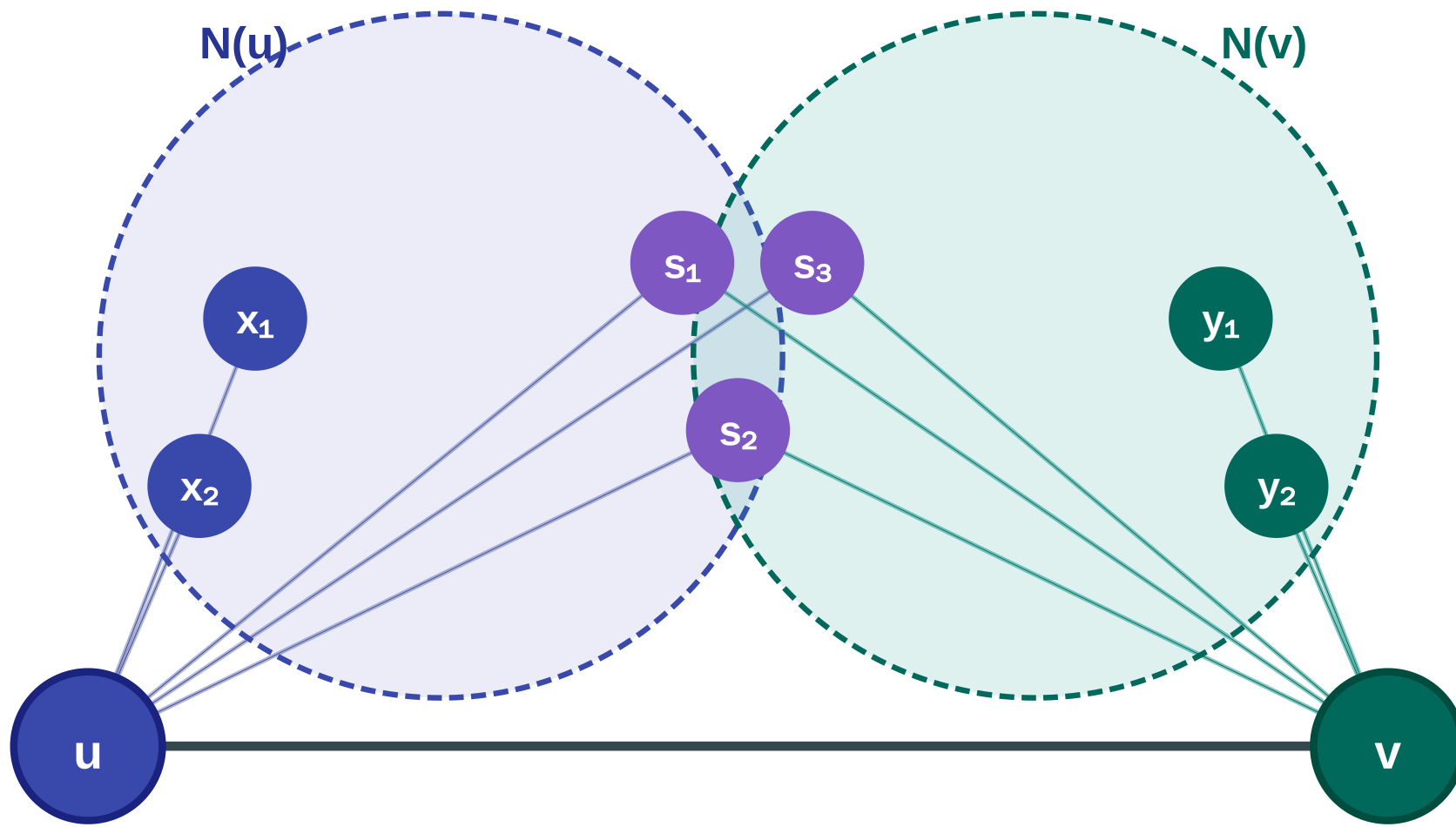
**Why this works as a heuristic.** Under STC, if  $v$  has many strong ties, those ties' endpoints must all be mutually connected — i.e.  $v$ 's neighborhood is dense. Conversely, a high  $C_v$  indicates a neighborhood that *could* be labeled mostly Strong without violating STC.  $C_v$  is thus a lower-bound signal for "how strong-embedded is  $v$ ".

# Neighborhood Overlap

**Neighborhood overlap.** For an edge  $(u, v)$ :

$$O(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}$$

where  $N(u)$  and  $N(v)$  exclude each other.  $O = 0$  identifies a pure local bridge;  $O = 1$  identifies an edge embedded inside a complete clique.



# Why These Heuristics Instead of Exact Labels

|                          | Exact MaxSTC  | Clustering coeff.    | Overlap           |
|--------------------------|---------------|----------------------|-------------------|
| Scope                    | whole graph   | per node             | per edge          |
| Complexity               | NP-hard       | $O(k^2)$ per node    | $O(k)$ per edge   |
| Captures                 | true tie type | neighborhood density | edge embeddedness |
| Feasible on real graphs? | No            | Yes                  | Yes               |

Here  $k$  is the **degree** of the node (or of an edge's endpoint) — i.e. its number of neighbors. Clustering coefficient checks all  $\binom{k}{2}$  pairs of a node's neighbors; overlap intersects the two endpoints' neighbor sets.

# Takeaway

Clustering coefficient and neighborhood overlap exist because MaxSTC is intractable. They are polynomial-time structural proxies for a theoretical object we cannot compute — trading precision for feasibility on real-world networks.

# Quick Check

Node  $B$  has 4 neighbors:  $A, C, D, E$ . Among  $B$ 's neighbors, exactly 2 edges exist:  $(A, C)$  and  $(C, D)$ .

1. Compute  $C_B$ .
2. Under STC, what is the maximum number of strong ties  $B$  can have?
3. What does  $C_B$  tell you about  $B$ 's structural role that STC alone does not?

# Quick Check — Solution

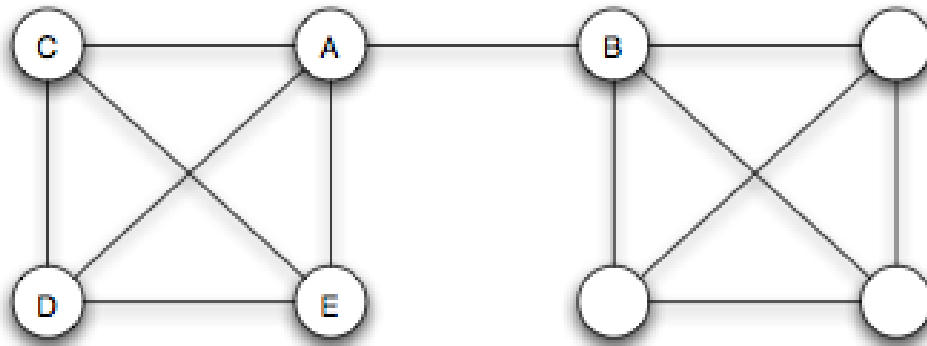
1.  $\binom{4}{2} = 6$  possible edges, 2 actual  $\Rightarrow C_B = 1/3$ .
2. Maximum strong ties from  $B$  is 2. The only pairs of  $B$ 's neighbors connected by an edge are  $(A, C)$  and  $(C, D)$ ; a third strong tie would force a currently missing edge (e.g.  $A-D$ ), violating STC.
3. The clustering coefficient is a *continuous* structural measurement —  $C_B = 1/3$  tells us  $B$  sits in a moderately sparse neighborhood, suggesting a potential brokerage position between the sub-clusters  $A, C$  and  $D$ . STC by itself gives only a hard upper bound on the number of strong ties but no information about the *shape* of the neighborhood. The heuristic sacrifices exactness for expressive power.

# Bridges, Local Bridges, and the Weak-Tie Theorem

**Guiding Question:** Which edges are structurally most important for keeping information flowing across a network — and can we predict from theory which ones those will be?

# Bridge

**Bridge.** An edge  $e \in E$  is a *bridge* if removing it increases the number of connected components. Equivalently:  $e$  lies on no cycle.

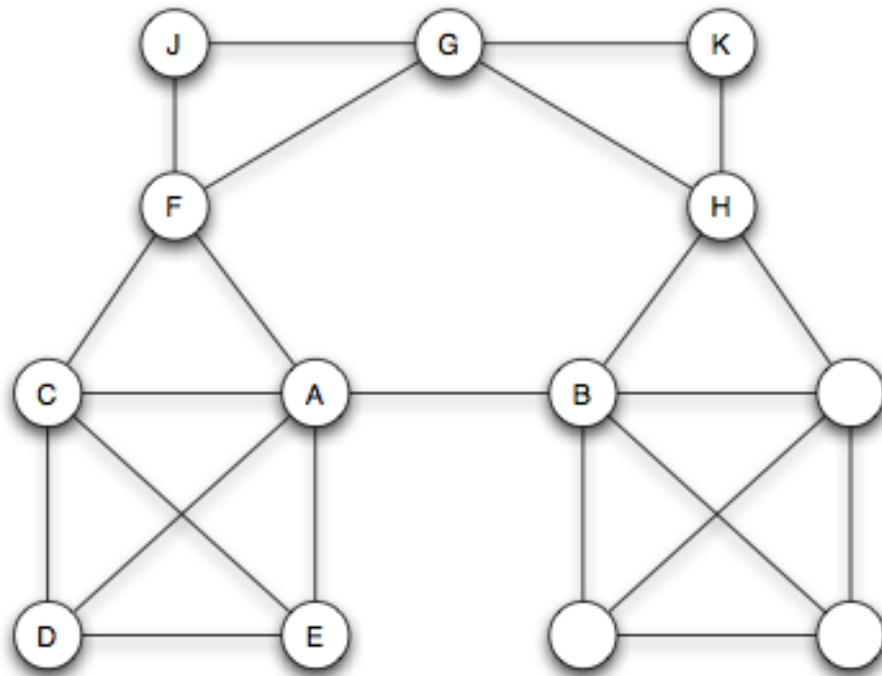


Source: Easley & Kleinberg 2010

Removing the highlighted edge splits the graph in two. The endpoints' distance becomes infinite — they can no longer reach each other through any other route.

# Local Bridge

**Local bridge.** An edge  $(u, v)$  is a *local bridge* if  $u$  and  $v$  have no neighbors in common:  $N(u) \cap N(v) = \emptyset$ . Equivalently:  $O(u, v) = 0$ .

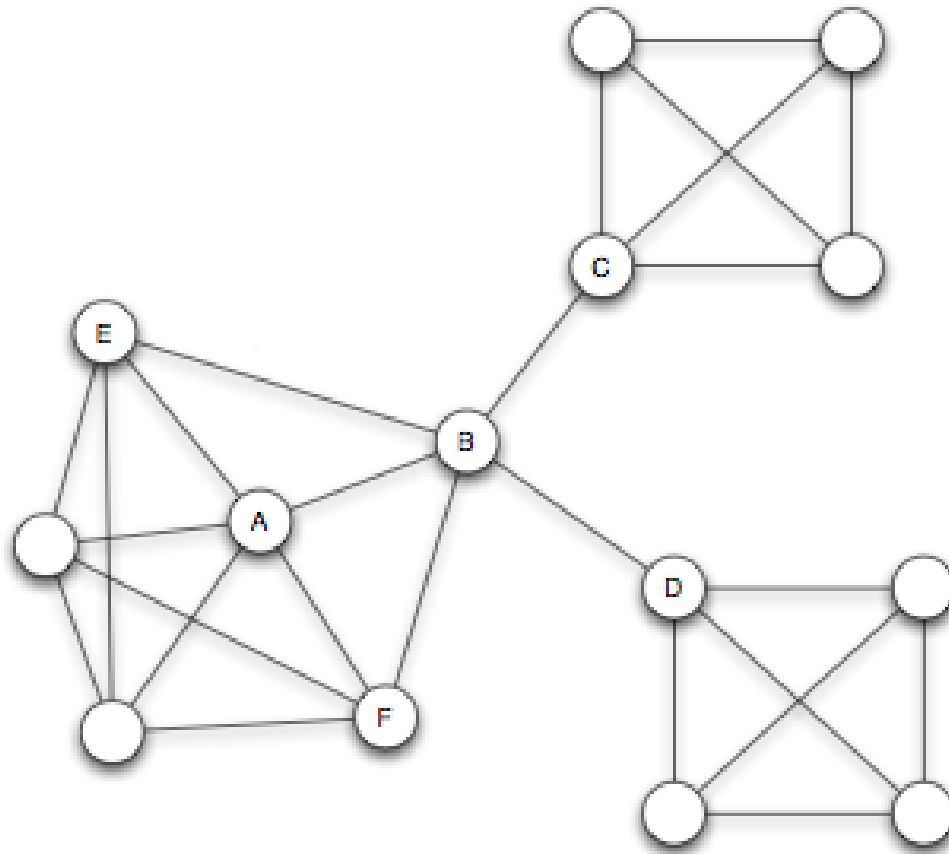


Source: Easley & Kleinberg 2010

A local bridge does not need to disconnect the graph — its endpoints may still reach each other via a long detour. Note that *local bridge* is already expressible in terms of the neighborhood-overlap heuristic: an edge with  $O = 0$  is exactly a local bridge.

# Structural Holes

A node that acts as the sole short link between otherwise separated groups spans a *structural hole* (Burt, 1992).



Source: Burt 1992

- **Information advantage:** the broker receives early, non-redundant signals from independent groups.
- **Control advantage:** the broker mediates flows between groups that cannot communicate directly.

# The Weak-Tie Theorem

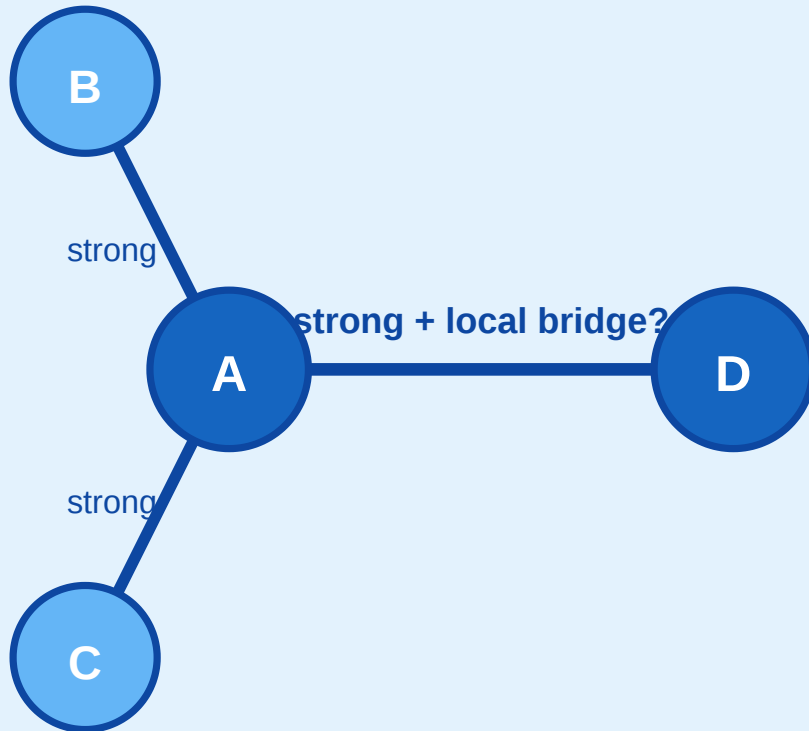
**Theorem (Granovetter).** Let  $\ell$  be a labeling of  $G$  that satisfies Strong Triadic Closure. Then every local bridge is labeled Weak under  $\ell$ .

This is the theoretical prediction we will test. It links the labeling model (STC, tie types) to a measurable structural property (local bridges) via an if-then statement.

# Proof Sketch by Contradiction

**Assume: A–D is strong AND a local bridge**

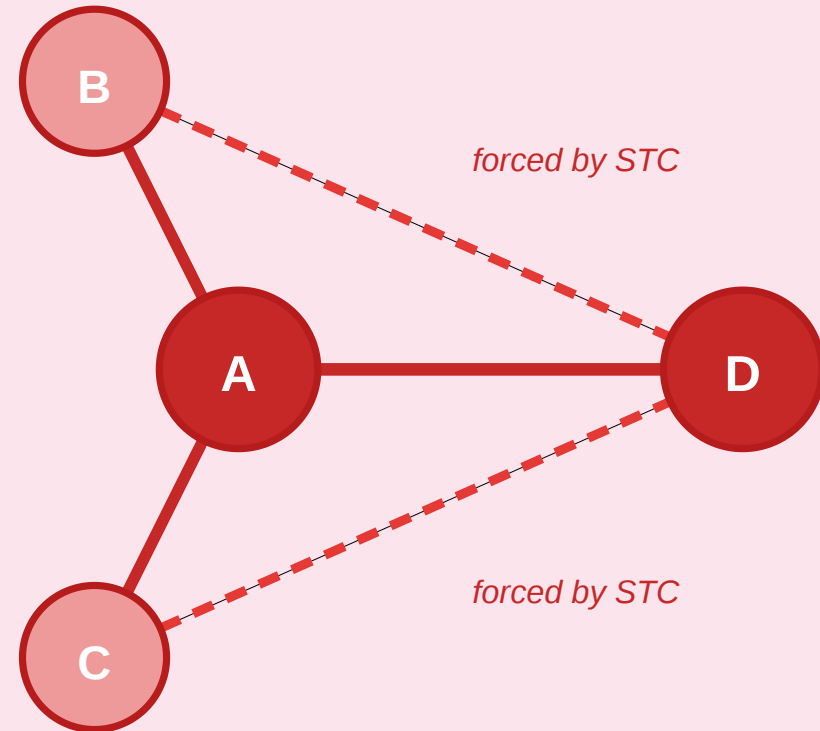
A also has strong ties to B and C



→  
STC forces

**Contradiction**

B and C must connect to D → shared neighbors!



A–D no longer a local bridge — B, C are shared neighbors

Assume  $A-D$  is both **strong** and a **local bridge**.  $A$  also has strong ties to  $B$  and  $C$ . By Strong Triadic Closure,  $D$  must be connected to  $B$  and  $C$ . But then  $B$  and  $C$  are shared neighbors of  $A$  and  $D$  — destroying the local-bridge condition. Contradiction.

# Takeaway

Under STC, local bridges must be weak ties. This is the theoretical prediction that connects our unobservable labeling model to the measurable neighborhood-overlap heuristic — and it is the prediction we now test against real data.

# Quick Check

A network has three nodes:  $A$ ,  $B$ ,  $C$ . The edges  $A-B$  and  $A-C$  are both labeled Strong. There is no edge between  $B$  and  $C$ .

1. Does this labeling satisfy Strong Triadic Closure?
2. Is  $A-B$  a local bridge? Does the Weak-Tie Theorem apply here?

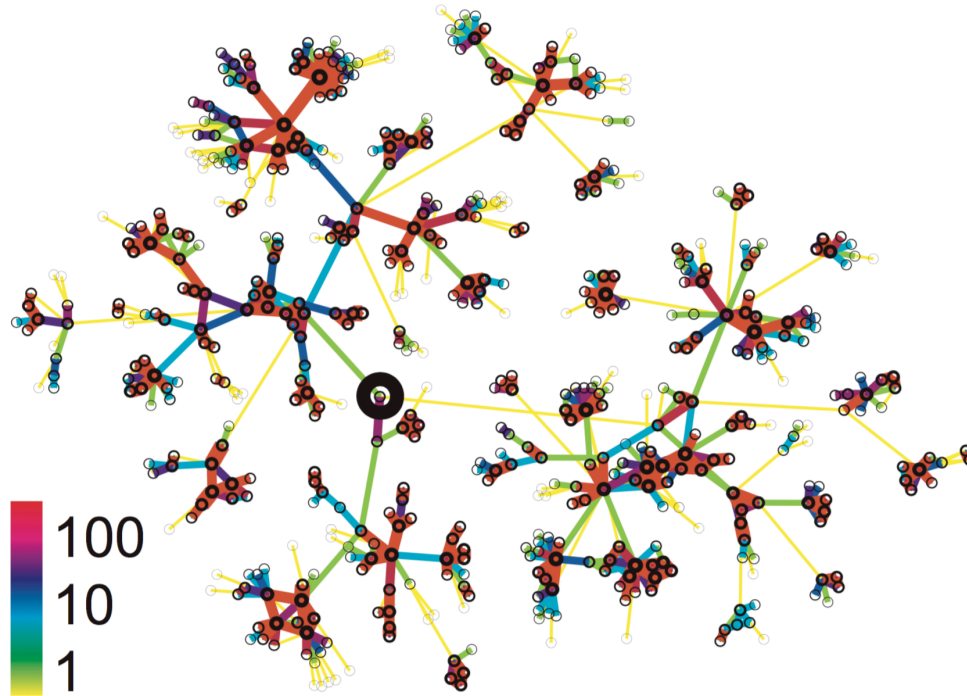
# Quick Check — Solution

1. No.  $A$  has two strong neighbors,  $B$  and  $C$ , but  $(B, C) \notin E$ . STC is violated.
2.  $A$  and  $B$  share no common neighbors in this graph, so  $A-B$  is technically a local bridge. But the Weak-Tie Theorem does *not* apply here: its precondition (STC holds) is violated. The theorem makes no prediction when its assumption fails.

# Empirical Validation at Scale

**Guiding Question:** Does the theoretical prediction — low overlap correlates with low tie strength — survive contact with real-world data at scale?

# Setup: Cell-Phone Networks

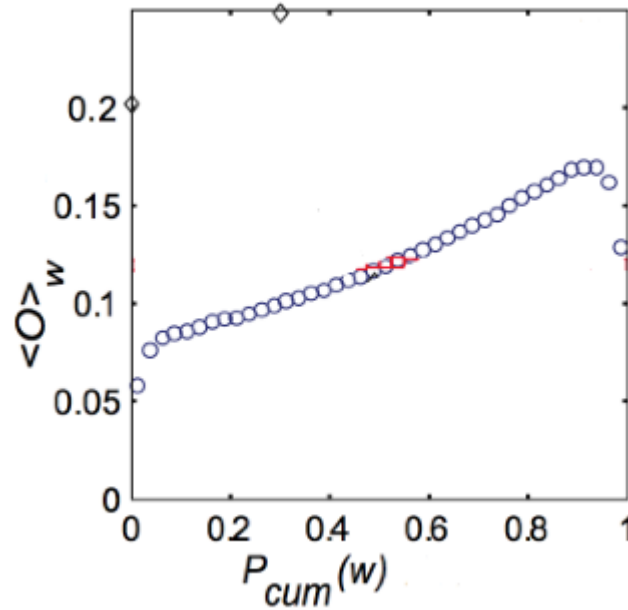


Source: Onnela et al. 2007

**Onnela et al. (2007).**

Nationwide cell-phone call logs modeled as an undirected weighted graph. Nodes are phone numbers; edge weight is the cumulative call duration between two numbers — a measurable proxy for tie strength.

# Evidence: Overlap vs. Tie Strength



Source: Onnela et al. 2007

X-axis: tie strength percentile (call duration). Y-axis: neighborhood overlap. Weak ties cluster near

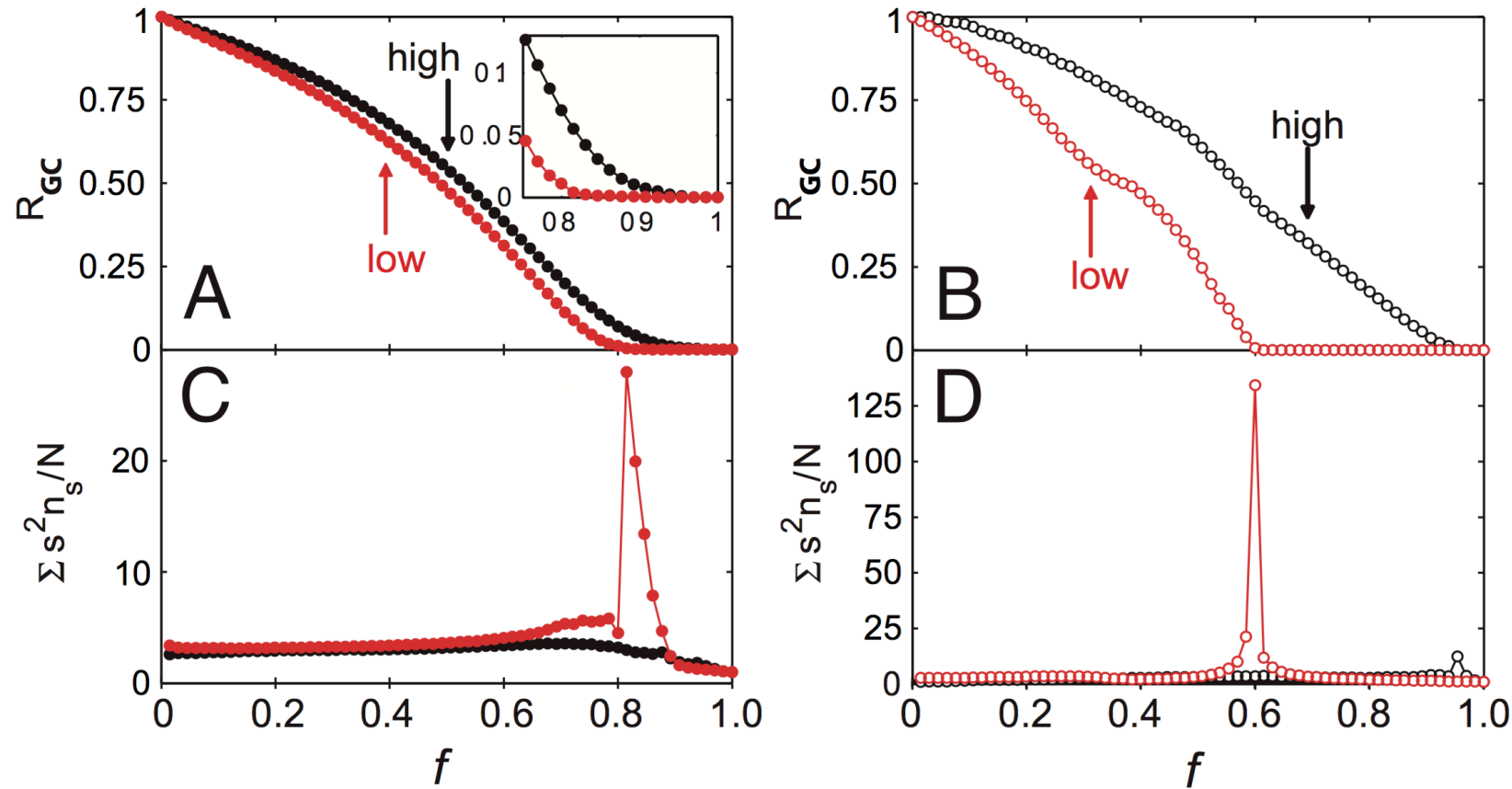
$O \approx 0$

; strong ties accumulate at higher

$O$

. The systematic monotonic relationship is exactly what the weak-tie theorem predicts.

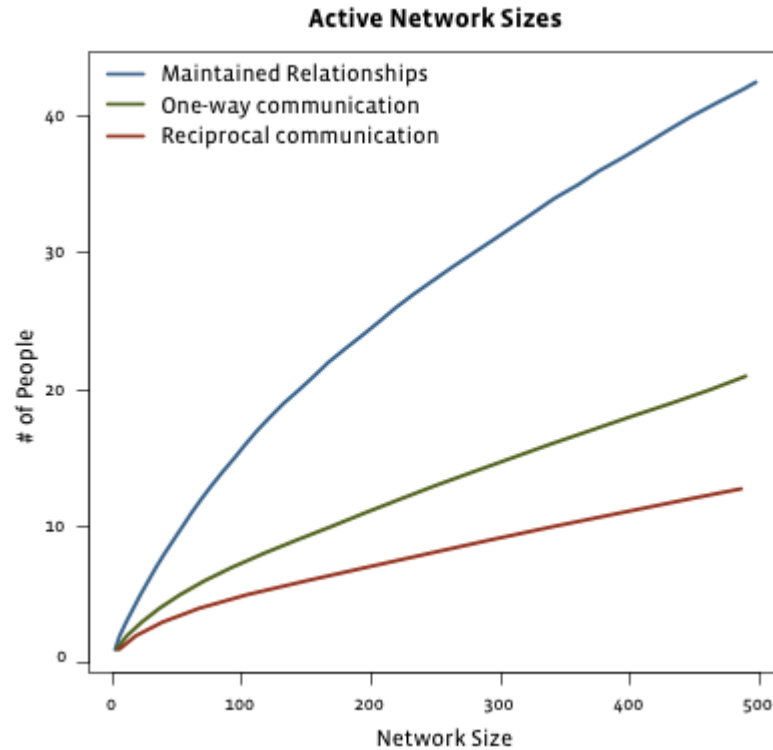
# Evidence: Network Robustness



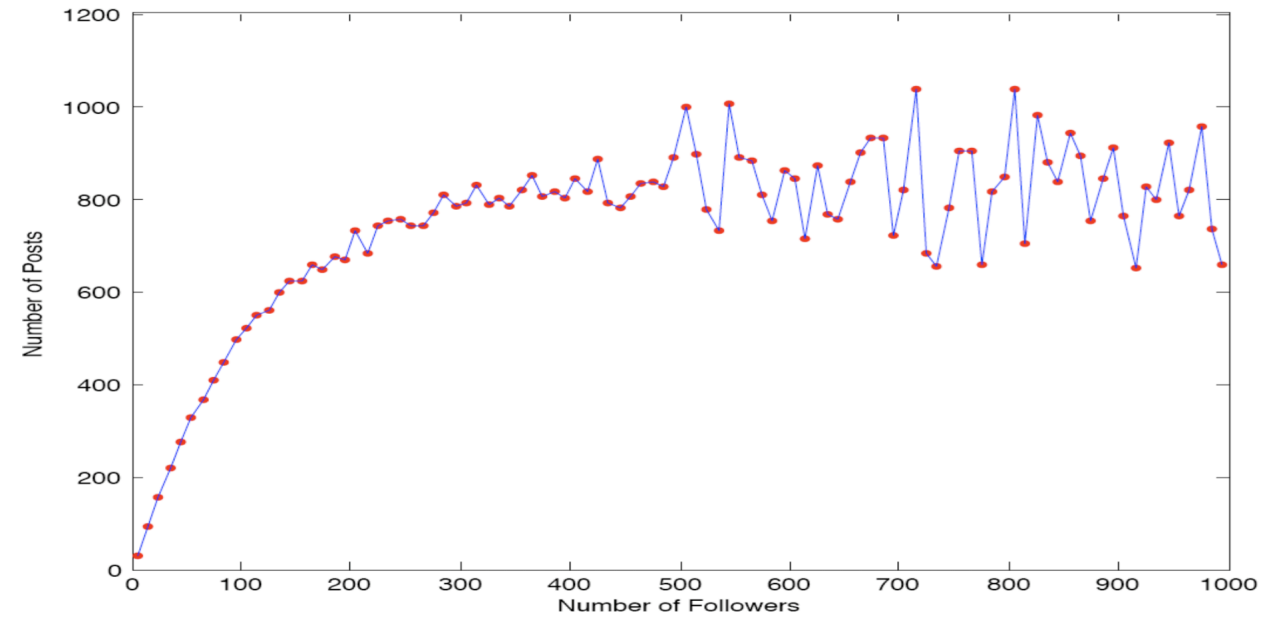
Source: Onnela et al. 2007

Removing edges in **increasing** tie-strength order (weakest first) fragments the giant component far faster than removing strongest ties first. Weak ties carry the structural connectivity of the whole network.

# Evidence: Social Media



Source: Easley & Kleinberg 2010



Source: Huberman et al. 2009

**Facebook:** total friends grow unboundedly, but *maintained* reciprocal ties plateau around Dunbar's number (~150).  
**Twitter:** followers scale freely, mutual interacting friends do not. Online platforms remove the cost of weak ties but do not change the cognitive bound on strong ones — amplifying the weak/strong distinction rather than collapsing it.

# Closing the Loop

We set out to test a theoretical model (typed edges + STC) on data where the true labels are unobservable and the exact recovery problem is NP-hard. The strategy was:

1. Define polynomial-time structural heuristics (clustering coefficient, overlap)
2. Derive a prediction from the theory (weak-tie theorem)
3. Measure the heuristics on real data and compare

The Onnela and social-media results close the loop: the heuristics behave exactly as the theorem predicts. The model earns its keep even though we could never compute its exact solution.

# Takeaway

The theoretical model (typed edges, STC, weak-tie theorem) cannot be inferred exactly from data, but its prediction — low overlap edges are weak — is confirmed at scale by independent empirical proxies. Model and measurement close the loop.

# Quick Check

Two students, X and Y, both use the same online platform. X follows 2000 accounts; Y follows 80 accounts but everyone follows everyone else.

1. Using neighborhood overlap as a proxy, whose connections look more "strong-tie-like"?
2. What does the weak-tie theorem predict about the diversity of information each student receives?

# Quick Check — Solution

1. Y's connections look more **strong-tie-like**: dense mutual follow cluster → high average neighborhood overlap. X's large following includes many algorithm-driven follows with near-zero overlap → low average, bridge-like.
2. The weak-tie theorem predicts:
  - X receives diverse, non-redundant information — structural signature of many weak, low-overlap edges acting as local bridges.
  - Y receives redundant, reinforced information — structural signature of a dense clique with few outgoing shortcuts.

The prediction matches the everyday pattern of algorithm-curated feeds vs. closed-friend echo chambers.

# Wrap-Up

- **Modeling** gave us triadic closure, typed edges, STC, and the weak-tie theorem
- **Measurement** forced us to abandon the exact labeling (NP-hard) and adopt polynomial-time proxies: clustering coefficient and neighborhood overlap
- **The theorem** provided the prediction that connects the two
- **Empirical data** at scale (Onnela, Facebook, Twitter) confirm the prediction

## Looking Ahead: Lecture 04

If ties reveal themselves through structural position, what about *groups*? How do we formally identify communities, and which nodes occupy structurally important roles inside and between them?

## Reading

- Chapter 3 of Easley & Kleinberg (2010).
- Kossinets & Watts, *Empirical Analysis of an Evolving Social Network*, [Science 311, 88–90 \(2006\)](#). — longitudinal evidence for triadic closure on a university email network.
- Sintos & Tsaparas, *Using Strong Triadic Closure to Characterize Ties in Social Networks*, KDD 2014 — NP-hardness of MaxSTC and tractable special cases.



## Image Credits & References

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Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning About a Highly Connected World. *Cambridge University Press*. <https://www.cs.cornell.edu/home/kleinber/net-works-book/>

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Onnela, J.-P. and Saramaki, J. and Hyvonen, J. and Szabo, G. and Lazer, D. and Kaski, K. and Kertész, J. and Barabási, A.-L. (2007). Structure and tie strengths in mobile communication networks. *Proceedings of the National Academy of Sciences*.